



Lichong Decoction represses colorectal cancer progression via inhibiting the PINK1/Parkin pathway regulated by Rab27B

Yu Dai^{a,1}, Fei Li^{a,1}, Yao Tang^a, Lingling Sun^a, Yongpu Liu^a, Jinyan Zhan^b,
Chuying Zhou^{b,*}, Aimin Li^{a,**}

^a Southern Medical University Hospital of Integrated Traditional Chinese and Western Medicine, Southern Medical University, Guangzhou, 510315, Guangdong, China

^b School of Traditional Chinese Medicine, Department of Traditional Chinese Medicine, Nanfang Hospital, Southern Medical University, Guangzhou, 510515, Guangdong, China

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ABSTRACT

Ethnopharmacological relevance: Lichong Decoction (LCD), a classical herbal formula, has been widely applied in clinical practice. Studies have validated its effectiveness in combating cancer; nevertheless, the exact pathways through which it influences colorectal cancer require further investigation.

Aim of the study: The mechanism of LCD in inhibiting colorectal cancer progression was investigated.

Materials and methods: BALB/c-*nu* mice were xenografted with colorectal cancer cells to establish an animal model. The chemical constituents of LCD were characterised using high-performance liquid chromatography. Quantitative real-time PCR was employed to assess the expression levels of Rab27B mRNA. The Cancer Genome Atlas data of patients with colorectal cancer were used for gene clinical correlation analysis. Lentiviral transfection was used to construct stable cell lines. Cell motility was analysed using wound-healing and invasion assays. Pathological changes were observed using haematoxylin and eosin staining. Protein expression was determined using immunohistochemistry and western blotting. The results were statistically analysed using GraphPad Prism 10.1.2.

Results: *In vivo*, LCD treatment led to a substantial decrease in the size of colorectal cancer tumours. Importantly, LCD markedly lowered the expression of Rab27B and proteins related to the PINK1/Parkin pathway in tumour tissues. *In vitro*, LCD significantly reduced the expression of Rab27B in colorectal cancer cells. Consistent with the results *in vivo*, the reduction in Rab27B inhibited PINK1/Parkin signalling protein expression and prevented the migration and invasion of colorectal cancer cells.

Conclusions: Lichong Decoction inhibits colorectal cancer growth by dampening PINK1/Parkin signalling in colorectal cancer cells through repressing Rab27B.

1. Introduction

Colorectal cancer (CRC) ranks among the most prevalent malignancies globally and is a leading contributor to cancer-related morbidity and mortality (Eng et al., 2022), accounting for roughly 7 % of all newly diagnosed cancer cases and around 11 % of cancer-related deaths. Importantly, in almost all regions of the world, the incidence of CRC has increased in people born in the early 1950s due to the introduction of many environmental chemicals (Murphy and Zaki, 2024). By 2030, the

CRC burden is expected to increase by more than 60 % (Arnold et al., 2017; Young et al., 2024). Considering the carcinogenic effect of chemical factors, traditional Chinese medicine, as a natural compound, has been widely used in the treatment and prevention of CRC due to its safety, low toxicity, and minimal adverse reactions (Cao et al., 2024). However, the anti-tumour mechanism of traditional Chinese medicine must be further elucidated experimentally (Zou et al., 2024).

Various natural compounds have been identified as capable of promoting cancer cell apoptosis and demonstrating antitumor effects by

* Corresponding author.

** Corresponding author.

E-mail addresses: dy_smu@126.com (Y. Dai), lfei19@smu.edu.cn (F. Li), tangyao0626@126.com (Y. Tang), sl_smu@126.com (L. Sun), lyp_smu@sina.com (Y. Liu), zjy_smu@126.com (J. Zhan), zchuying_tcm@163.com (C. Zhou), liaimin2005@163.com (A. Li).

¹ These authors contributed equally to this work and shared first authorship.

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regulating mitophagy controlled by the PTEN-induced kinase 1 (PINK1)/Parkin pathway (Narendra and Youle, 2024). The most critical event in mitophagy is the formation of a mitochondrial autophagosome and mitochondria-derived vesicles (MDVs), which subsequently fuse with lysosomes to promote the degradation of mitochondrial content (Okamoto, 2014; Lemasters, 2014; Lu et al., 2023). The biogenesis of autophagosomes and lysosomes, which is activated by the PINK1/Parkin pathway, is regulated by mTORC1 (Nezich et al., 2015). Moreover, PINK1-mediated AKT activation influences mitochondrial function, and the induction of mitophagy is correlated with elevated PI3K/AKT and mTOR activities (Van Humbeeck et al., 2011; Lou et al., 2020). Rab27B has been implicated in multiple vesicular formation and trafficking events, including those involved in autophagosome and lysosome processes (Ostrowski et al., 2010; Underwood et al., 2020; Chen et al., 2024). In addition, previous research has demonstrated that Rab27B modulates cytokine secretion within cells, thereby affecting the activity of the PI3K/AKT and mTORC1 pathways (Dai et al., 2021). Rab27B belongs to the Rat Sarcoma (RAS) oncogene family, and mutated RAS activation has been shown to facilitate mitophagy (Ren et al., 2023). Increasing evidence links Rab27B to the development of various cancers, including colorectal cancer (Cheng et al., 2019). However, whether Rab27B regulates the PINK1/Parkin signalling pathway and thus affects tumour progression has not yet been studied.

Lichong Decoction (LCD), a classic Chinese herbal medicine from the *Records of Traditional Chinese and Western Medicine in Combination* written by Zhang Xichun, has been recommended for treating colorectal cancer (Liu et al., 2024). *Astragalus membranaceus* (Fisch.) Bunge., *Codonopsis pilosula* (Franch.) Nannf., *Atractylodes macrocephala* Koidz., *Callus Gallus Domesticus* Brisson, and *Dioscorea opposita* Thunb., which tonify qi and the spleen, constitute the primary components of LCD. These herbs significantly enhance the body's immune function, inhibit tumour progression, and promote recovery (Chen et al., 2016; Luan et al., 2021; X. Wei et al., 2024; Xu et al., 2024). Herbs with stasis-transforming properties, such as *Curcuma kwangsiensis* S.G. Lee & C.F. Liang and *Sparganium stoloniferum* (Buch.-Ham. ex Graebn.) Buch.-Ham. ex Juz., further contribute to the antitumor efficacy of LCD (Lin et al., 2024; Ni et al., 2025). Additionally, *Anemarrhena asphodeloides* Bge. and *Trichosanthes kirilowii* Maxim. nourish yin and clear heat, thereby improving the tumour microenvironment (Zhang et al., 2024; Zou et al., 2024). Although astragaloside, the primary active ingredient in *A. membranaceus*, has been reported to regulate mitochondrial homeostasis and autophagy processes (Wang et al., 2024), and curcumin, the main component of *C. kwangsiensis*, has demonstrated modulating the interrelated processes of mitochondrial renewal, autophagy, and apoptosis (Moustapha et al., 2015), the mechanism of action of LCD, a compound traditional Chinese medicine prescription, remains unclear. This study reveals that LCD potentially inhibits the progression of CRC by suppressing Rab27B-mediated PINK1/Parkin signalling.

2. Materials and methods

2.1. Drug preparation and bioactive compounds analysis of LCD

LCD consists of nine commonly used herbs (local name): *Dangshen*, *Baizhu*, *Huangqi*, *Shanyao*, *Jinei jin*, *Sanleng*, *Zhimu*, *Ezhu*, and *Tianhuafen*. The corresponding botanical plant names and drug sources are shown in Tables 1 and 2. The plant names have been matched using <http://www.theplantlist.org>. The raw herbs were procured from the Southern Medical University Hospital of Integrated Traditional Chinese and Western Medicine (Guangzhou, China). All 87 g of raw herbs were mixed and boiled in 1000 ml of water three times for 20 min each. The resulting extract was subsequently concentrated to a final concentration of 2.262 g/mL using a rotary evaporator. The primary active constituents of LCD was checked using a Shimadzu high-performance liquid chromatography (HPLC) system consisting of an LC-20A binary pump, SIL20A autosampler, and a CTO-20A column oven. Chromatographic separation

Table 1
The drug composition list of Lichong Decoction (LCD).

Local name	Botanical plant names	The dosage used (Raw herbs dried)	Parts used
Dangshen	<i>Codonopsis pilosula</i> (Franch.) Nannf.	6g	Root
Baizhu	<i>Atractylodes macrocephala</i> Koidz.	6g	Rootstock
Huangqi	<i>Astragalus membranaceus</i> (Fisch.) Bunge.	9g	Root
Shanyao	<i>Dioscorea opposita</i> Thunb.	15g	Rootstock
Jinei jin	<i>Gallus gallus domesticus</i> Brisson	9g	Gizzard lining
Sanleng	<i>Sparganium stoloniferum</i> (Buch.-Ham. ex Graebn.) Buch.-Ham. ex Juz.	9g	Tuber
Zhimu	<i>Anemarrhena asphodeloides</i> Bge.	12g	Rootstock
Ezhu	<i>Curcuma kwangsiensis</i> S.G.Lee & C.F.Liang	9g	Rootstock
Tianhuafen	<i>Trichosanthes kirilowii</i> Maxim.	12g	Root

Table 2
The source of various drugs in Lichong Decoction (LCD).

Local name	Name of manufacturer	Product lot number
Dangshen	Daxiang Pharmaceutical Group Co., Ltd. Guangdong China	HX23L01
Baizhu	Guangdong Herbal Tablets Factory. Guangdong China	B4272211
Huangqi	Anhui Fuzhong Pharmaceutical Co., Ltd. Anhui China	220903
Shanyao	Foshan Pharmaceutical Materials Co., Ltd., China National Medicine Group Fengliaoxing. Guangdong China	C22312251
Jinei jin	AhGuoYuan Traditional Chinese Medicine Co., Ltd. Anhui China	22092501
Sanleng	Guangzhou Kangsheng Pharmaceutical Co., Ltd. Guangdong China	210901
Zhimu	Guangdong Herbal Tablets Factory. Guangdong China	Z2322913
Ezhu	Guangdong Zhongtian Pharmaceutical Co. Ltd. Guangdong China	20220401
Tianhuafen	Foshan Pharmaceutical Materials Co., Ltd., China National Medicine Group Fengliaoxing. Guangdong China	C22301053

was performed using a ShimNex CS C18 column (I.D. 4.6 mm × 250 mm, 5 μm, Japan). The mobile phase consisted of solvents A (water) and B (acetonitrile). A 40-min run with gradient elution is shown in Table 3. The flow rate was set at 1 ml/min, and the column temperature was set to 35–40 °C. Cisplatin (0.5 mg/ml) and Fluorouracil (2.5 mg/ml) were obtained from Nanfang Hospital and dissolved in 0.9 % saline for use in animal experiments. Most of the Standard Chinese medicine ingredients were purchased from Shanghai Standard Technology Co., Ltd. (Supplementary Material 1), while a small portion was purchased from Sichuan Vikeqi Biotechnology Co., Ltd (Supplementary Material 2). All aforementioned solutions were stored at −20 °C until use.

Table 3
Gradient elution system used for HPLC analyses of LCD.

Time (min)	Mobile phase A (%)	Mobile phase B (%)
0.1	95	5
1	95	5
10	80	20
35	30	70
40	25	75

2.2. Generation of stable cell line

Target cell infection was performed according to the manufacturer's instructions (GenePharma, Suzhou, China). SW480 cell suspension was prepared at a density of 0.5×10^5 cells/well, seeded in 24-well plates, and incubated overnight. On the second day, Rab27B-594, Rab27B-697, Rab27B-778, LV5-NC, LV3-NC, Rab27B-OE, or HiTransG A infection reagents were added according to the manufacturer's instructions. After 48 h of culture, fresh medium was added. The infection efficiency was recorded 72 h after infection using fluorescence microscopy. The stable transfected cell lines were screened with an optimal concentration of 2 $\mu\text{g/mL}$ of puromycin. The transfection efficiency was determined using western blotting (Xiao et al., 2018).

2.3. Cell culture

Cell lines were sourced from the American Type Culture Collection (ATCC), and the culture procedures described previously. Specifically, SW480-EGFP, MC38, RKO, and Caco-2 cells were cultured in Dulbecco's Modified Eagle's Medium (DMEM) enriched with 10 % (v/v) foetal bovine serum (FBS) and 1 % (v/v) penicillin-streptomycin. SW480, DLD-1, and LoVo cells were cultured in Roswell Park Memorial Institute medium (RPMI-1640), also supplemented with 10 % FBS and 1 % penicillin-streptomycin. 1 $\mu\text{g/mL}$ puromycin was used for the maintenance culture of all stable cell lines. All cell lines were cultured at 37 °C in a humidified atmosphere of 95 % air and 5 % CO₂.

2.4. CRC model

Five-to six-week-old BALB/c-*nu* mice were sourced from the Laboratory Animal Center of Southern Medical University (Guangdong Experimental Animal Quality Certification Number: 44002100035868). Orthotopic tumours were established by implanting tumour slices into the mesenteric region, 1 cm proximal to the caecal terminus. These tumour slices were obtained from BALB/c-*nu* mice that had been previously xenografted with EGFP-labelled SW480 cells. The animals were housed in a sterile environment and randomly allocated to various experimental groups: Model group (200 μL , 0.9 % saline kg/d, intra-gastric administration), LCD low-dose group (LD; 5.65 g/kg/d), LCD medium-dose group (MD; 11.31 g/kg/d), LCD high-dose group (HD; 22.62 g/kg/d), and cisplatin plus fluorouracil group (DDP + Fu, cisplatin, 5 mg/kg twice a week and fluorouracil 50 mg/kg once a week, intraperitoneal injection). The 11.31 g/kg dosage was calculated based on the clinical prescription dosage for humans ($87 \text{ g}/70 \text{ kg} = 1.24 \text{ g/kg}$), accounting for the body surface area differences between humans and animals ($1.24 \text{ g/kg} \times 9.1 = 11.31 \text{ g/kg}$) (Gou et al., 2023). Upon completion of the experiment, the mice were humanely euthanised, and their respective tumour, liver, and spleen tissues were harvested for further analysis. All animal research procedures adhered to the Guidelines for the Care and Use of Laboratory Animals established by the National Institutes of Health. The experimental protocols received approval from the Animal Experiment Ethics Committee at Southern Medical University (ethical approval number: SMUL2023026; March 10, 2023).

2.5. The Cancer Genome Atlas clinical data analysis

The gene expression (Supplementary Material 3) and clinical files (Supplementary Material 4) of patients with CRC were downloaded from The Cancer Genome Atlas (TCGA) official repository, accessible at <https://portal.gdc.cancer.gov/>, and the R programming language was used to analyse the data and create graphs. The detailed analysis code for the R programming language was sourced from the data mining course of the Bilibili website <https://www.bilibili.com/>.

2.6. Western blotting analysis

Total protein was extracted and quantified using a Bicinchoninic Acid (BCA) Protein Assay Kit (Epizyme Biomedical Technology, Shanghai, China). For western blot (WB) analysis, primary antibodies were used at various dilutions, including Rab27B (1:1000; Proteintech Group, Chicago, IL, USA, Cat No. 13412-1-AP), PINK1 (1:1000; Proteintech Group, Cat No. 23274-1-AP), Parkin (1:2000; Proteintech Group, Cat No. 14060-1-AP), LC3B (1:500; Proteintech Group, Cat No. 14600-1-AP), and β -Actin (1:5000; Proteintech Group, Cat No. 20536-1-AP). Following incubation with the primary antibodies, horseradish peroxidase (HRP)-conjugated secondary antibodies (diluted 1:10000) were applied for 1 h at room temperature after washing with TBST. The immunoreactive protein bands were visualised using an Enhanced Chemiluminescence system (Bio-Rad, Hercules, CA, USA) and subsequently imaged.

2.7. Immunohistochemistry staining

Tissues were processed for immunohistochemistry (IHC) following the manufacturer's protocols. Tissue sections were gently washed with phosphate-buffered saline and subsequently incubated overnight at 4 °C with primary antibodies against Rab27B (1:100), PINK1 (1:200), Parkin (1:100), and LC3B (1:200). On the following day, the sections were incubated with corresponding secondary antibodies (Zhongshan Golden Bridge, Beijing, China) and counterstained with haematoxylin before mounting. Positive staining areas for Rab27B, PINK1, Parkin, and LC3B were quantified using the ImageJ software. Protein expression levels in IHC-stained sections were evaluated using the H-score formula: H-score = percentage of positive area (as measured by ImageJ) $\times$ intensity score. Intensity scores were assigned based on the degree of staining: light, moderate, and deep.

2.8. Haematoxylin-eosin staining

Four micrometre thick slices of each tumour tissue, dehydrated and embedded in paraffin, were prepared as unstained versions by an experienced pathologist from the Southern Medical University Hospital of Integrated Traditional Chinese and Western Medicine. The slices were stained with haematoxylin solution for 3–5 min after dewaxing. The slices were soaked in haematoxylin differentiation solution for 1–2 s. The slices were blued with Scott Tap Bluing. After each step of staining, the slices were rinsed with running water. The slices were dehydrated and sealed with neutral glue. Images were acquired and analysed with a microscope.

2.9. Quantitative real-time PCR

Total RNA was isolated from tissue samples using the Total RNA Isolation Kit (Foregene, Chengdu, China), following a procedure that included cell lysis, RNA extraction, precipitation, and purification through washing steps. The extracted RNA was then reverse transcribed into cDNA using the Real-Time RT Reagent Kit (TakaRa, Kusatsu, Japan) according to the manufacturer's protocol. Relative mRNA expression levels were quantified by normalising to the endogenous control gene glyceraldehyde-3-phosphate dehydrogenase (GAPDH) using the comparative C(T) (2^{-($\Delta\Delta\text{CT}$)}) method. Custom primers were synthesised by Sangon Biotech (Shanghai, China), with the sequences listed in Table 4.

Table 4

Primer sequences used in this study.

Gene	Forward primer (5' $\rightarrow$ 3')	Reverse primer (5' $\rightarrow$ 3')
GAPDH	CCAGGGCTGCTTTTAACCTCT	GGACTCCACGACGTACTCA
Rab27B	GACACTGCGGGACAAGAGC	CTTGCGGTTCATTGACTTCC

2.10. Wound healing, cell migration, and invasion assay

Eight micrometre-thick pore inserts (Corning, USA) to 24-well culture plates. Stable cell lines that overexpressed or knocked down Rab27B cells (0.5×10^5 cells) were placed in the lower chambers, and SW480 cells were placed in the upper chambers and co-cultured in RPMI 1640 with 2 % FBS for 48 h. Matrigel gum was used as the substrate for the invasion assay. Cells overexpressing or knocked down for Rab27B were plated in 6-well plates and incubated overnight to achieve approximately 80 % confluence. A scratch with a width of approximately 1 mm was created on the cell monolayer using the tip of a 200- μ L pipette, and images of the scratches were captured at 0 and 48 h post-

scratch. All experiments were conducted in triplicate.

2.11. Statistical analyses

Statistical analyses were performed using GraphPad Prism 10.1.2 (GraphPad Software) (Guo et al., 2024). Each experiment was carried out at least in triplicate, and the results were averaged and presented as the mean $\pm$ standard deviation (SD). To assess the significance of differences between two independent groups, a two-tailed Student's t-test was applied, with * indicating $p < 0.05$, ** indicating $p < 0.01$, and *** indicating $p < 0.001$.

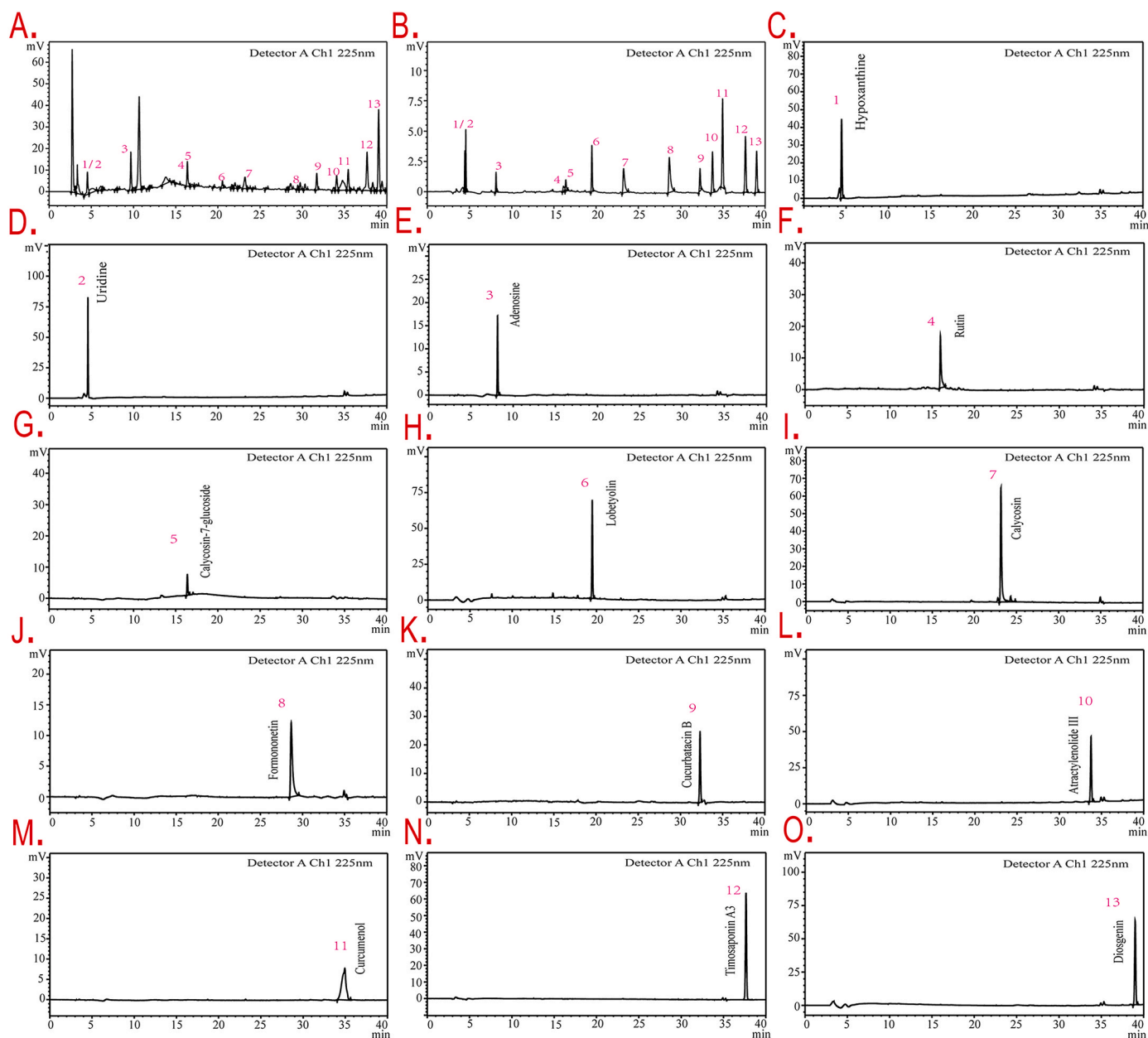


Fig. 1. Qualitative evaluation of LCD.

(A) HPLC fingerprints of LCD. (B) Thirteen representative ingredients of LCD. (C) Hypoxanthine (effective components of Jineijin). (D) Uridine (effective components of Huangqi). (E) Adenosine (effective components of Dangshen, Huangqi, Shanyao). (F) Rutin (effective components of Sanleng and Huangqi). (G) Calycosin-7-glucoside (effective components of Dangshen and Huangqi). (H) lobetyolin (effective components of Dangshen). (I) Calycosin (effective components of Dangshen and Huangqi). (J) Formononetin (effective components of Sanleng, Dangshen, and Huangqi). (K) Cucurbitacin B (effective components of Tianhuafen). (L) Atractylenolide III (effective components of Baizhu). (M) Curcumenol (effective components of Ezhu). (N) Timosaponin A3 (effective components of Zhimu). (O) Diosgenin (effective components of Shanyao).

3. Results

3.1. Detection of the main effective components of LCD

HPLC is the primary method used to determine the quality of Chinese medicinal compounds (Li et al., 2011). It can quickly separate and detect various components of Chinese medicines, ensuring the quality and safety of the drugs. HPLC analysis was employed to initially assess the quality of the LCD mixture (Supplementary Material 5), and the HPLC profiles confirmed that *Hypoxanthine*, *Uridine*, *Rutin*, *Lobetyolin*, *Cucurbitacin B*, *Attractylenolide III*, *Curcumenol*, *Timosaponin A3*, and *Diosgenin*—effective components of *Jineijin*, *Huangqi*, *Sanleng*, *Dangshen*, *Tianhuaifen*, *Baizhu*, *Ezhu*, *Zhimu*, and *Shanyao*, respectively—were present in the mixture (Fig. 1, Supplementary Material 6). Notably, various Chinese herbs may share identical active constituents. For instance, both *Dangshen* and *Huangqi* contain *Adenosine*, *Calycosin-7-glucoside*, *Calycosin*, and *Formononetin*.

3.2. Anticancer activity of LCD in vivo

In the *in-vivo* experiment, the tumours removed from mice in the HD and DDP + Fu groups were markedly smaller compared to those in the model group (Fig. 2A). Statistical evaluation demonstrated a significant reduction in tumour size due to LCD treatment ($p < 0.001$; Fig. 2B). These results indicated that the high-dose LCD group significantly inhibited the progression of CRC. Critically, compared to the DDP + Fu group, the LCD-treated mice did not exhibit significant weight loss (Fig. 2C). Haematoxylin and eosin (H&E) staining demonstrated that the spleen colour and size remained normal in LCD-treated mice, with no evident liver inflammation or injury (Fig. 2D). Collectively, these results indicate that LCD exhibits robust antitumor activity in mice without observable adverse effects on body weight, liver, or spleen health.

3.3. Rab27B expression inhibited by high-dose LCD

High Rab27B expression has been described as a marker of poor prognosis in several cancers, including CRC (Afroz et al., 2024). The results from the TCGA database analysis also showed that Rab27B is closely related to CRC progression (Fig. 3). To investigate whether the anticancer effects of LCD were linked to Rab27B expression, Rab27B levels in tumours were assessed using western blotting, immunohistochemistry, and quantitative real-time PCR. The findings revealed that Rab27B expression was markedly upregulated in the model group but significantly reduced in the high-dose group. These analyses confirmed that high-dose LCD treatment led to a substantial decrease in Rab27B expression in mouse tumour tissues (Fig. 4). These results indicate that the anti-CRC efficacy of LCD is associated with decreased Rab27B expression.

3.4. PINK1/Parkin signalling protein expression was inhibited by high-dose LCD

Increasing evidence has shown that mitophagy influences cancer stemness, immune cell infiltration, radioresistance, and chemotherapy resistance. PINK1 and Parkin are critical regulators of the classical mitophagy pathway. Western blotting and immunohistochemistry were employed to assess the expression of marker proteins related to PINK1/Parkin signalling. Western blot analysis revealed that the expression levels of LC3B (a marker for mitochondrial autophagy), PINK1, and Parkin were elevated in the control group but markedly reduced in the high-dose group (Fig. 5A and B). These findings were corroborated by IHC analysis (Fig. 5C and D). Collectively, these results indicate that the anti-CRC effects of LCD are linked to the inhibition of PINK1/Parkin signalling.

3.5. Rab27B regulates PINK1/Parkin signalling to affect the invasion ability of CRC cells

Afroz et al. identified Rab27B as a critical component involved in the regulation of autophagy in CRC. Mitochondrial autophagy, regulated by PINK1/Parkin signalling, is a selective autophagy process that plays a crucial role in maintaining cell homeostasis and human health. A high dose of LCD was used to treat three different CRC cell lines with high Rab27B expression: SW480, DLD-1, and MC38. Similarly, significant inhibition of Rab27B expression by LCD was observed in CRC cells (Fig. 6A and B). Rab27B overexpression and knockdown stable cell lines were constructed (Fig. 6C and D), and the expression of proteins involved in PINK1/Parkin signalling assessed using western blot analysis. Compared to the control group, the results indicated that LC3B, PINK1, and Parkin expression were significantly upregulated by Rab27B overexpression (Rab27B-OE) but downregulated by Rab27B knockdown (Rab27B-778; Fig. 7A and B). Additionally, the migration and invasion capabilities of CRC cells were evaluated through wound healing and invasion assays. Overexpression of Rab27B significantly improved the migration efficiency of CRC cells by approximately 87.8 %; however, after Rab27B knockdown, the migration rate was only 25.3 % (Fig. 7C and D). These results indicate that inhibition of Rab27B expression can significantly restrain PINK1/Parkin signalling and thus decrease the invasion and migration of tumour cells.

4. Discussion

CRC is one of the most common malignancies worldwide and has a high mortality rate. According to the latest World Health Organization data, CRC accounts for 9.4 % of cancer-related deaths worldwide (Kanth and Inadomi, 2021). The incidence of CRC tends to increase with changes in dietary habits and the chemical environment. Surgical removal, chemotherapy, and radiation are the standard treatments; however, more than 30 % of patients with resectable CRC experience relapse (Biller and Schrag, 2021), highlighting that anticancer treatments still face serious challenges. Currently, fluorouracil drugs, such as 5-fluorouracil, are used to treat CRC, but their long-term use is limited due to adverse reactions (Grothey et al., 2021). Hence, it is crucial to develop safer and more effective methods to prevent the occurrence and progression of CRC.

LCD's safety and efficacy have been substantiated through more than a century of clinical applications. Research has demonstrated that the incorporation of *Huanglian*, *Huangbai*, and *Baihua Sheshe cao* into LCD effectively inhibits the Wnt/ β -catenin signalling pathway and epithelial-mesenchymal transition, thereby mitigating the progression of CRC (Liu et al., 2024). The intricate composition of traditional Chinese medicinal components contributes to their multi-target antitumor effects. *Dangshen* extract in LCD inhibits autophagy and induces apoptosis of colon cancer cells via inhibiting the expression of LC3B and Beclin-1 (Luan et al., 2018). *Huangqi* crude extracts inhibit the growth of CRC *in vivo* by regulating PI3K/Akt, mTOR, and other pathways (Wang et al., 2019; Y. Wei et al., 2024). *Baizhu* inhibits extracellular vesicle metastasis in CRC (Tang et al., 2020; Jiang et al., 2022). *Zhimu* and *Ezhu* exhibit significant anti-tumour activity and have been utilized in treating CRC (Kang et al., 2011; Sun et al., 2024). These components (Fig. 1) may underlie the potential anticancer mechanisms of LCD. In the present study, LCD served as a tumour-suppressing formulation for colorectal cancer treatment by inhibiting Rab27B, thus deactivating the PINK1/Parkin signalling pathway and restraining tumour growth in xenograft models without inducing toxic side effects (Fig. 2). Nevertheless, this study remains qualitative, and further in-depth investigation is required for a comprehensive quantitative analysis of the anticancer constituents in LCD.

High Rab27B expression is an unfavourable prognostic factor in many solid cancers (Koh et al., 2020). Hendrix et al. found that Rab27B promoted breast cancer proliferation, aggressiveness, and tumour

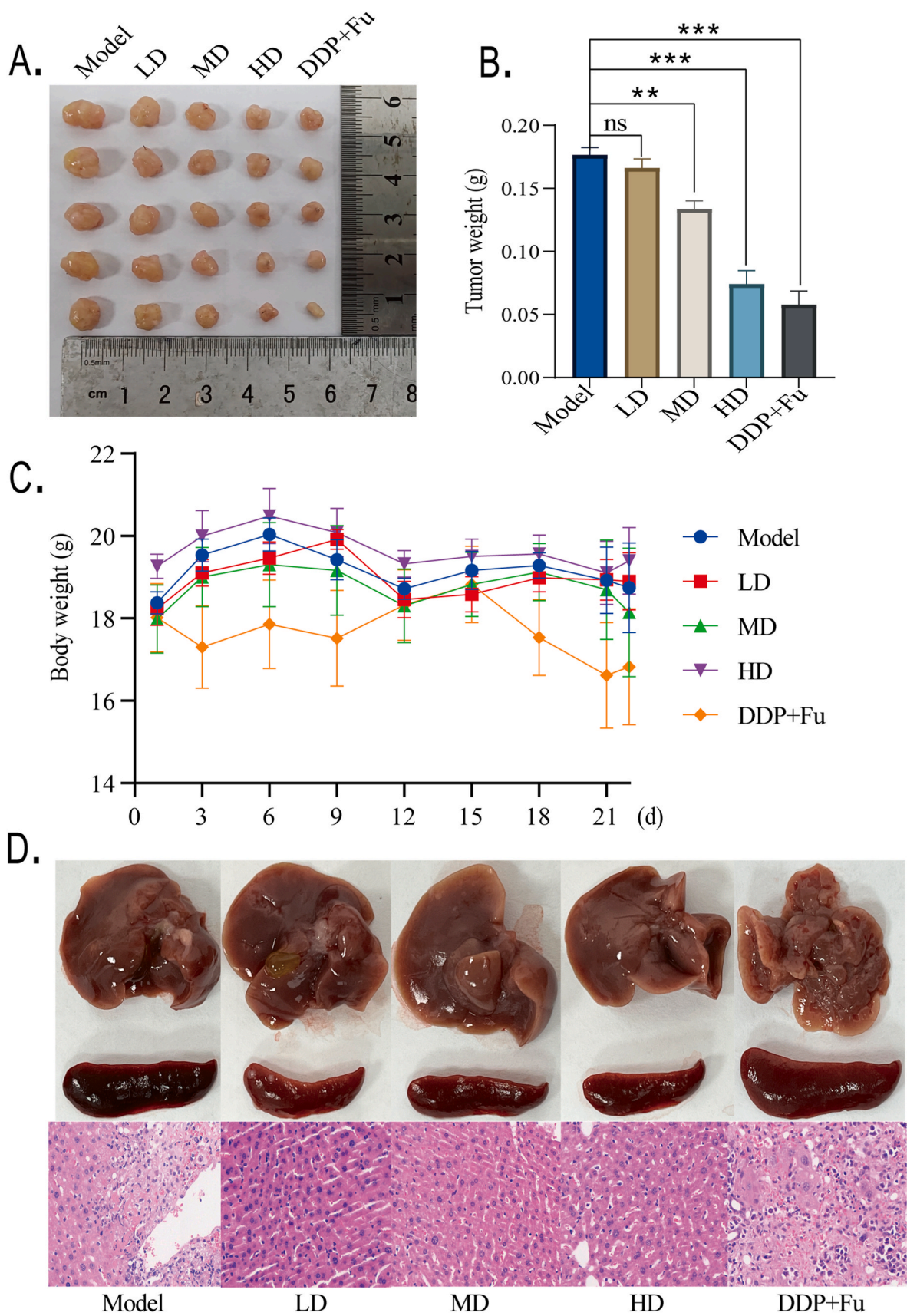


Fig. 2. LCD inhibits the growth of CRC. (A) Photograph of the tumour excised from a mouse. (B) Tumour weight and (C) body weight of mice. (D) Photograph of the liver and spleen excised from mice and H&E staining of the liver. Data are presented as the mean $\pm$ SD ($n = 5$); *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

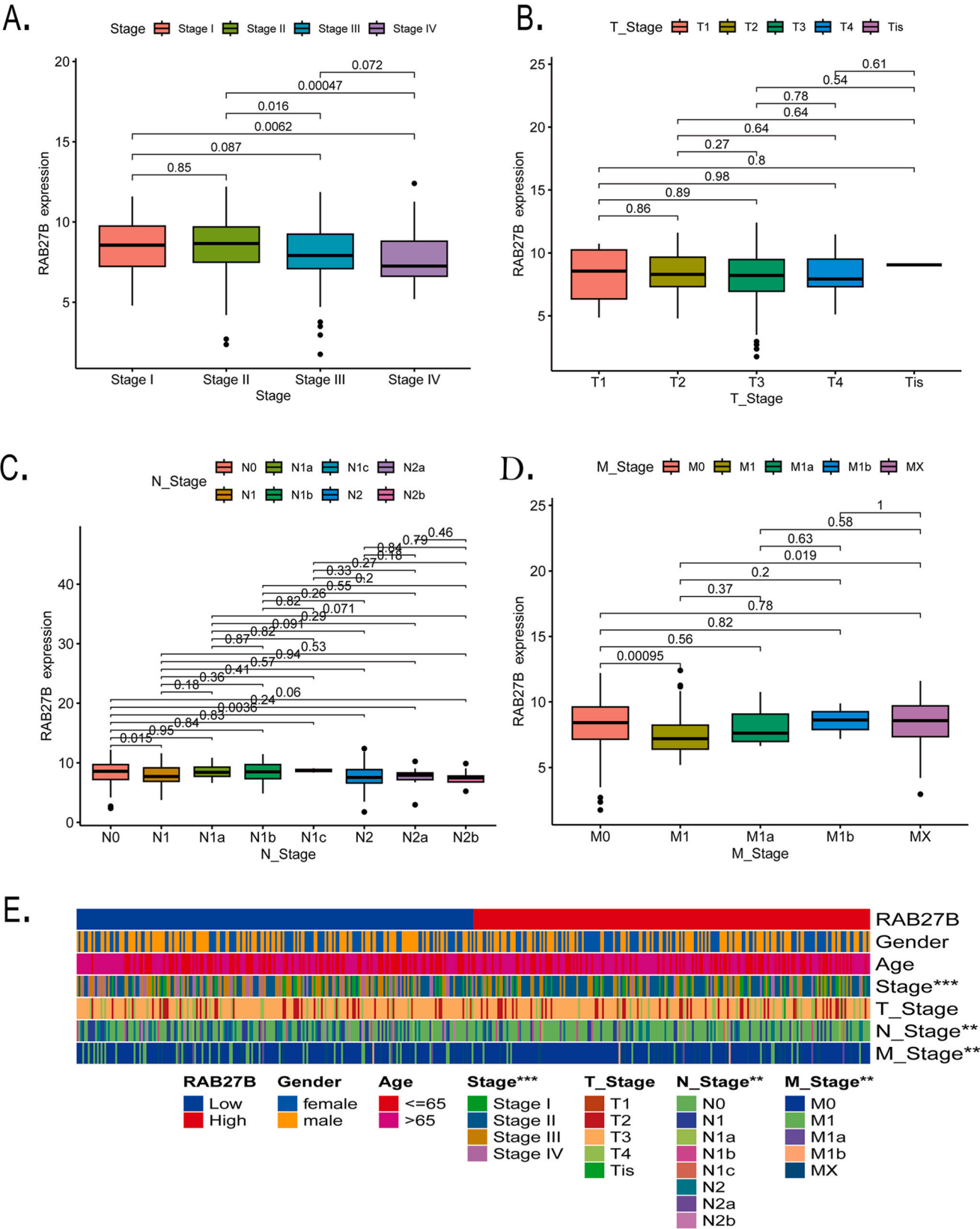


Fig. 3. Clinical correlation analysis of Rab27B expression in patients with colorectal cancer. (A) Correlation analysis between Rab27B expression and stage. (B) Correlation analysis between Rab27B expression and T-stage. (C) Correlation analysis between Rab27B expression and N-stage. (D) Correlation analysis between Rab27B expression and M-stage. (E) Heat map of clinical correlation analysis of Rab27B expression in patients with colorectal cancer. Data are presented as the mean $\pm$ SD; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

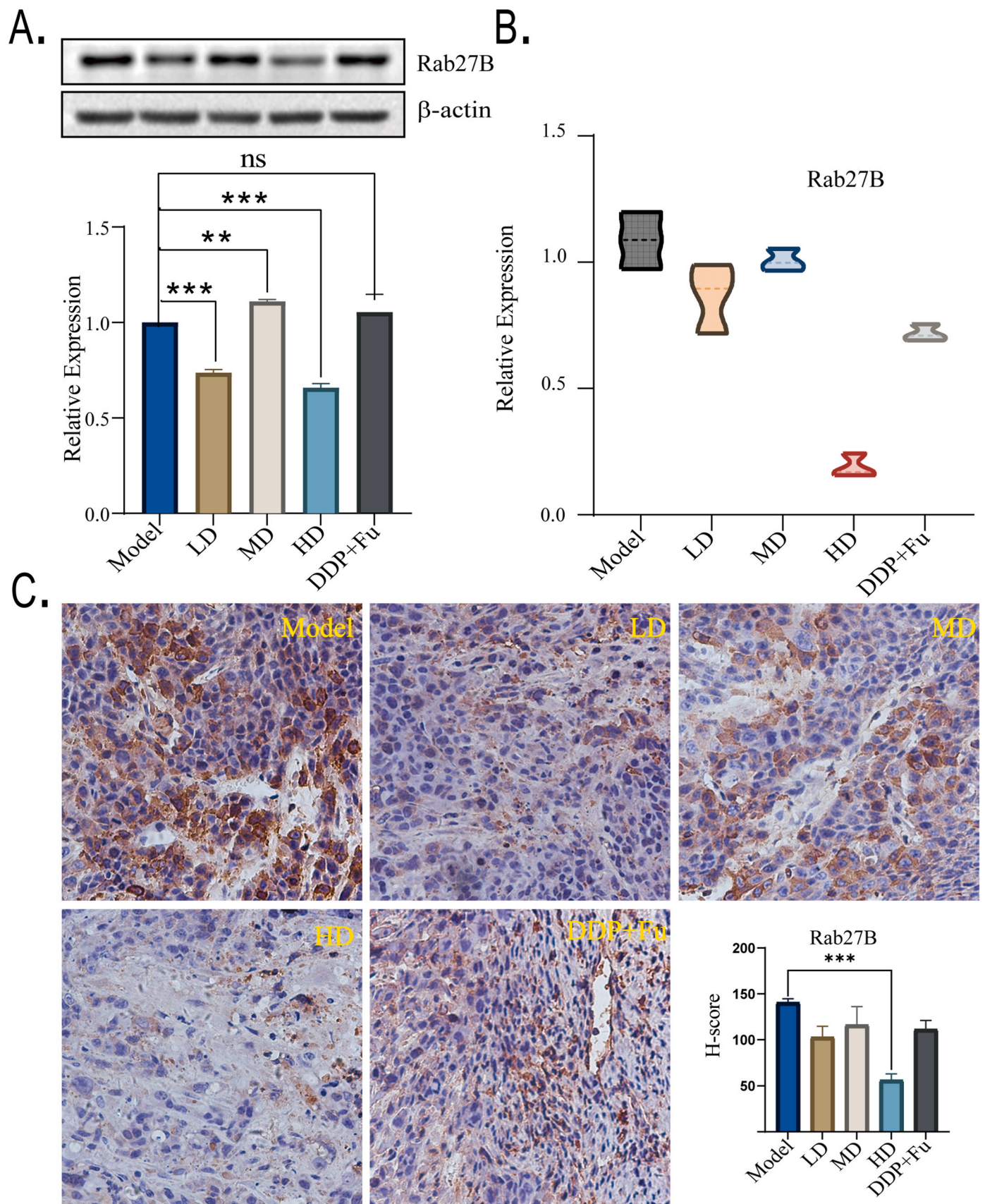


Fig. 4. LCD suppresses the expression of Rab27B in colorectal cancer.

(A) Western blot analysis of Rab27B and relative expression of Rab27B protein. (B) The mRNA expression level of Rab27B was detected using qPCR. (C) The expression of Rab27B was evaluated by immunohistochemistry and quantitative analysis of Rab27B expression levels. Data are presented as the mean $\pm$ SD; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

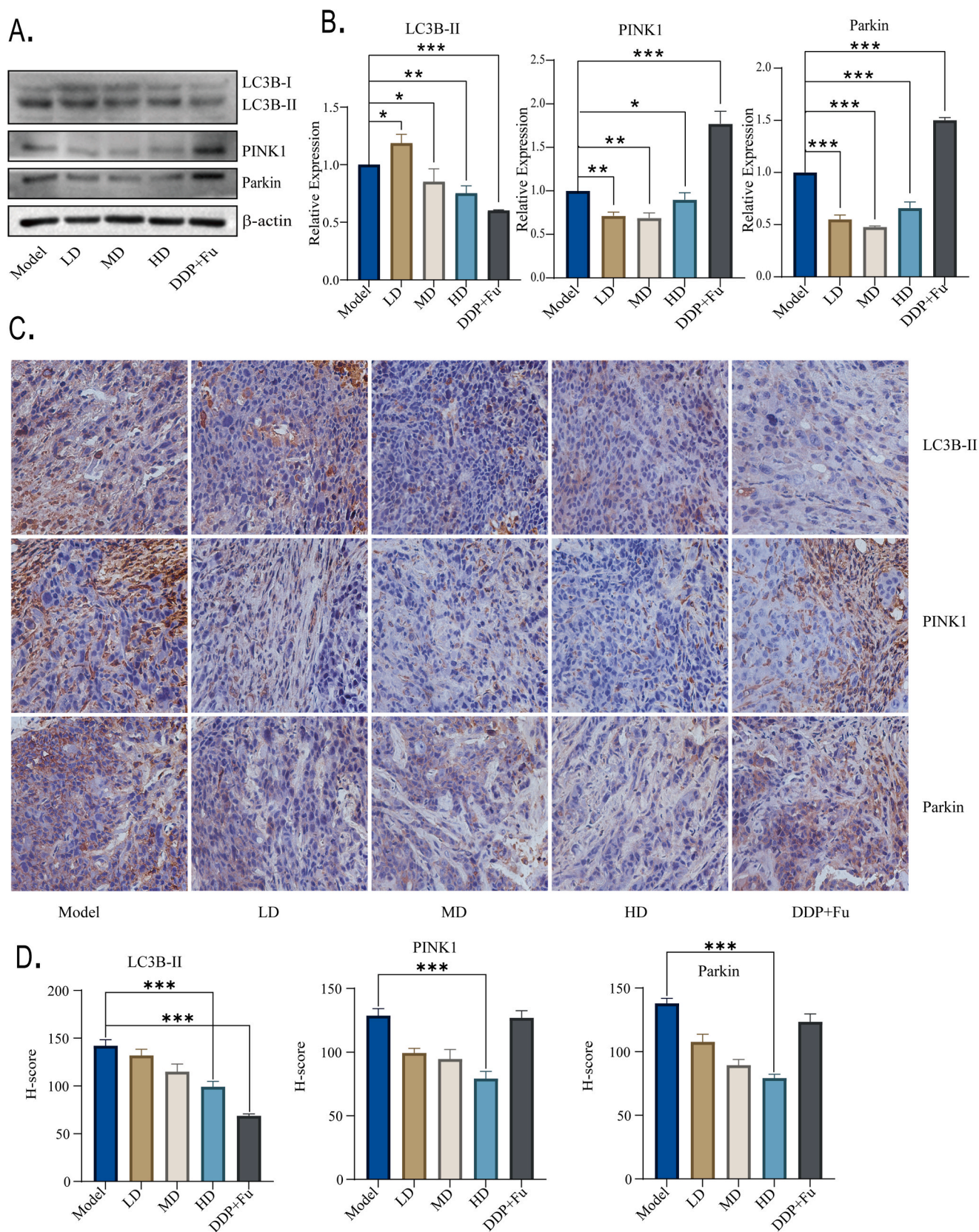


Fig. 5. LCD suppresses the expression of PINK1/PARKIN signalling pathway proteins in colorectal cancer.

(A) Western blot analysis of LC3B, PINK1, and Parkin. (B) Relative analysis of LC3B, PINK1, and Parkin protein expression levels. (C) Expression of LC3B, PINK1, and Parkin was evaluated using immunohistochemistry. (D) Quantitative analysis of LC3B, PINK1, and Parkin expression levels based on immunohistochemistry. Data are presented as the mean $\pm$ SD; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

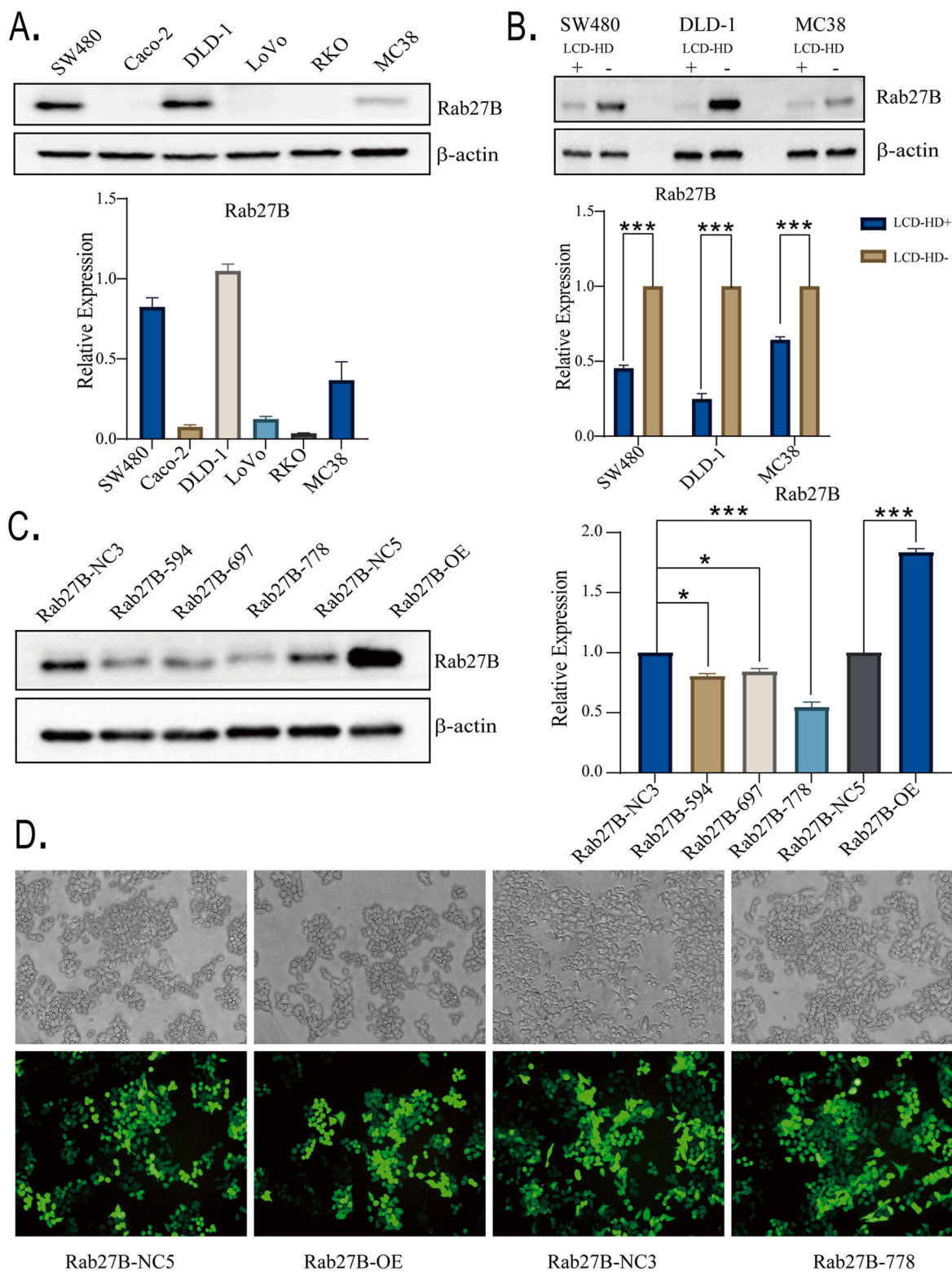


Fig. 6. LCD suppresses the expression of Rab27B in colorectal cancer cell lines.

(A) Western blot analysis and quantification of Rab27B protein expression in different colorectal cancer cell lines. (B) Western blot analysis and relative quantification of Rab27B after stimulation with LCD in three cell lines with high Rab27B expression. (C) Western blot analysis and relative quantification of Rab27B after lentiviral transfection of SW480 cells. (D) Immunofluorescence detection of cells stably overexpressing and with knocked-down Rab27B. Data are presented as the mean $\pm$ SD; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

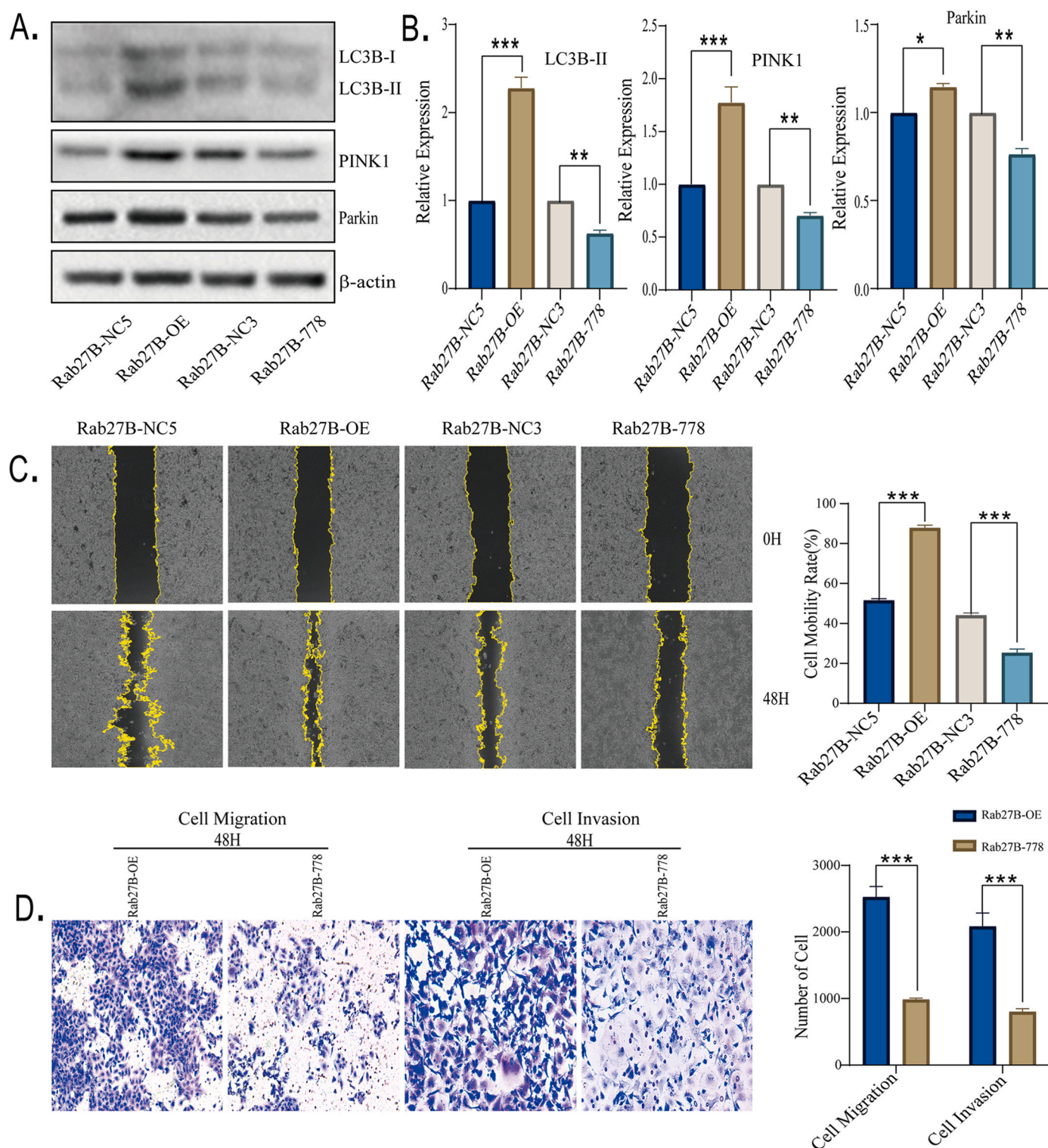


Fig. 7. Rab27B mediates the PINK1/Parkin signalling pathway and affects the migration and invasion ability of colorectal cancer cells. (A) Western blot analysis of LC3B, PINK1, and Parkin. (B) Relative analysis of protein expression levels of LC3B, PINK1, and Parkin. (C) Wound healing assay of stable cell lines. (D) Migration and invasion of stable cell lines. Data are presented as the mean $\pm$ SD; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

growth (Hendrix et al., 2010). Yuting et al. found that Rab27B induced extracellular vesicle secretion, endowing fibroblasts with the cancer-associated fibroblast phenotype, which enhanced the lymph node metastasis of bladder cancer (Li et al., 2023). Rab27B was identified as a potential predictor of CRC metastasis (Fig. 3). The results of the present study show that Rab27B was highly expressed in the CRC model

group (Fig. 4). Consistent with previous reports, knockdown of Rab27B in CRC cells resulted in reduced cancer cell migration and invasion (Fig. 7C and D). Therefore, an effective drug targeting Rab27B is a potential strategy for innovative CRC treatment. The results indicated that high-dose LCD significantly reduced the expression of Rab27B (Figs. 4, 6A and 6B), suggesting that LCD's anticancer effects on CRC are

associated with the downregulation of Rab27B. These observations highlight the complex role of Rab27B in various cancers and provide further evidence that Rab27B may serve as a promising therapeutic target for CRC treatment.

Rab27B, a member of the RAS oncogene family, plays a crucial role in regulating vesicle formation, transport, and exocytosis. Activating RAS mutations (*Harvey RAS*, Neuroblastoma RAS [*NRAS*], and Kirsten RAS [*KRAS*]) are carcinogenic drivers observed in up to 19 % of all human cancers (Jerchel et al., 2018). Among the RAS subtypes, mutations in the *KRAS* and *NRAS* have frequently been identified, and *KRAS* mutations in CRC are as high as 50 % (Biller and Schrag, 2021). Activated RAS induces the production of reactive oxygen species, thereby mediating mitochondrial autophagy. During mitophagy, PINK1, a serine/threonine kinase, functions with PRKN/parkin, an E3 ubiquitin ligase, to target damaged mitochondria to the lysosomes for degradation. Importantly, Rab27B upregulation activates PRKN/Parkin signalling, while the depletion of Rab27B suppresses oncogenic NRAS signalling (Ren et al., 2023). The current data (Figs. 5, 7A and 7B) showed that the suppressed expression of Rab27B downregulated PINK1/Parkin signalling protein and LC3B expression in CRC, further suggesting that LCD suppresses CRC by downregulating the Rab27B mediated PINK1/Parkin pathway.

Notably, within the context of a dynamic signalling environment, the regulatory mechanisms governing mitophagy are highly complex. To date, three distinct forms of mitophagy have been identified: type 1, which is mediated by PI3K; type 2, which relies on PINK1 and Parkin; and type 3, also known as micromitophagy, in which MDVs are selectively removed from the mitochondria and subsequently internalised into multivesicular bodies in a PINK1/Parkin-dependent manner (Lemasters, 2014). Previous studies have demonstrated that Rab27B modulates the activity of the PI3K/AKT pathway (Dai et al., 2021). In this current investigation, it was further identified that Rab27B regulates the expression of proteins involved in the PINK1/Parkin signalling pathway. Consequently, Rab27B may serve as a potential candidate molecule for regulating various forms of mitophagy. However, whether LCD exerts its broad regulatory effects through Rab27B remains to be elucidated in future research. Notably, this study specifically examined the effects of LCD on the Rab27B-mediated PINK1/Parkin signalling pathway.

PINK1/Parkin-mediated mitophagy is triggered by the loss of membrane potential and protein aggregation (Kitada et al., 1998). Polyubiquitin on the surface of impaired mitochondria recruits ubiquitin-binding autophagy adaptor proteins such as OPTN, NDP52, and LC3, among which LC3 is probably an important autophagy adaptor (Shi et al., 2014; Narendra and Youle, 2024). Mitophagy exhibits complex effects in cancer, and a more thorough understanding of the relationships between human cancer characteristics and alterations in PINK1/Parkin signalling is necessary to elucidate the clinical significance of targeting mitophagy in oncology (Li et al., 2022; Picca et al., 2023; Arcos et al., 2024). This study aimed to explore the functions and potential mechanisms of LCD in the development of colorectal tumours. The study demonstrated that LCD inhibited CRC tumourigenesis by targeting Rab27B-mediated PINK1/Parkin signalling. These findings provide novel antitumor strategies for the treatment of CRC. Additionally, the therapeutic potential of natural compounds in cancer treatment was examined by focusing on the PINK1/Parkin pathway. Overcoming these challenges could pave the way for the development of novel and innovative cancer therapies.

5. Conclusions

Rab27B-induced PINK1/Parkin signalling is critical for CRC progression. LCD inhibited CRC progression, and an optimal effect was observed at high doses. LCD inhibited CRC growth by downregulating PINK1/Parkin Signalling caused by Rab27B. Thus, LCD is an attractive therapeutic option for patients with CRC.

CRedit authorship contribution statement

Yu Dai: Writing – original draft, Validation, Methodology, Funding acquisition, Conceptualization. **Fei Li:** Visualization, Validation, Software. **Yao Tang:** Visualization, Methodology. **Lingling Sun:** Data curation. **Yongpu Liu:** Software. **Jinyan Zhan:** Formal analysis. **Chuying Zhou:** Supervision, Investigation. **Aimin Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Ethics statement

The study was approved by the Nanfang Hospital Animal Ethics Committee to the animal protocol, application No. L2023026.

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Declaration of competing interest

We the undersigned declare that this manuscript entitled “**Lichong Decoction represses colorectal cancer progression via inhibiting the PINK1/Parkin pathway regulated by Rab27B**” is original, has not been published before and is not currently being considered for publication elsewhere. All data were generated in-house, and no paper mill was used. All authors agree to be accountable for all aspects of work ensuring integrity and accuracy.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

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Abbreviations

CRC	colorectal cancer
LCD	Lichong Decoction
EGFP	enhanced green fluorescent protein
HPLC	high-performance liquid chromatography
PINK1	PTEN-induced Putative Kinase 1
LC3B	a subtype of microtubule-Associated Protein Light Chain 3
mTORC1	a subtype of mammalian Target Of Rapamycin Complex
PI3K	Phosphoinositide-3-Kinase
AKT	protein kinase B
MDVs	mitochondria-derived vesicles
WB	western blotting
DDP + Fu	cisplatin plus fluorouracil group;
HD	LCD high-dose group;
MD	LCD medium-dose group;
LD	LCD low-dose group;
ATCC	American Type Culture Collection
DMEM	Dulbecco's Modified Eagle's Medium
RPMI-1640	Roswell Park Memorial Institute medium
FBS	fetal bovine serum

TCGA	The Cancer Genome Atlas
BCA	Bicinchoninic Acid
GAPDH	glyceraldehyde-3-phosphate dehydrogenase
SD	standard deviation

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.119997>.

Data availability

Data will be made available on request.

References

- Afroz, S., Preet, R., Vishwakarma, V., Evans, A.E., Magstadt, A.N., Dixon, D.A., 2024. Regulation of autophagy by rab27b in colorectal cancer. *Int. J. Biochem. Cell Biol.* 177, 106693. <https://doi.org/10.1016/j.biocel.2024.106693>.
- Arcos, M., Goodla, L., Kim, H., Desai, S.P., Liu, R., Yin, K., Liu, Z., Martin, D.R., Xue, X., 2024. PINK1-deficiency facilitates mitochondrial iron accumulation and Colon tumorigenesis. *Autophagy* 1–17. <https://doi.org/10.1080/15548627.2024.2425594>.
- Arnold, M., Sierra, M.S., Laversanne, M., Soerjomataram, I., Jemal, A., Bray, F., 2017. Global patterns and trends in colorectal cancer incidence and mortality. *Gut* 66 (4), 683–691. <https://doi.org/10.1136/gutjnl-2015-310912>.
- Biller, L.H., Schrag, D., 2021. Diagnosis and treatment of metastatic colorectal cancer: a review. *JAMA* 325 (7), 669–685. <https://doi.org/10.1001/jama.2021.0106>.
- Cao, M., Tang, Y., Luo, Y., Gu, F., Zhu, Y., Liu, X., Yan, C., Hu, W., Wang, S., Chao, X., Xu, H., Chen, H., Wang, L., 2024. Natural compounds modulating mitophagy: implications for cancer therapy. *Cancer Lett.* 582, 216590. <https://doi.org/10.1016/j.canlet.2023.216590>.
- Chen, M., May, B.H., Zhou, I.W., Sze, D.M., Xue, C.C., Zhang, A.L., 2016. Oxaliplatin-based chemotherapy combined with traditional medicines for neutropenia in colorectal cancer: a meta-analysis of the contributions of specific plants. *Crit. Rev. Oncol. Hematol.* 105, 18–34. <https://doi.org/10.1016/j.critrevonc.2016.07.002>.
- Chen, Y., Wen, J., Li, Q., Peng, D., Liao, C., Ma, X., Wang, M., Niu, J., Wang, D., Li, Y., Zhang, X., Zhou, H., Zou, J., Li, L., Liu, L., 2024. RAB27b-regulated exosomes mediate LSC maintenance via resistance to senescence and crosstalk with the microenvironment. *Leukemia* 38 (2), 266–280. <https://doi.org/10.1038/s41375-023-02097-3>.
- Cheng, W., Liao, T., Lin, C., Yuan, L.E., Lan, H., Lin, H., Teng, H., Chang, H., Lin, C., Yang, C., Huang, S., Jiang, J., Yang, S., Yang, M., Hwang, W., 2019. RAB27b-activated secretion of stem-like tumor exosomes delivers the biomarker microRNA-146a-5p, which promotes tumorigenesis and associates with an immunosuppressive tumor microenvironment in colorectal cancer. *Int. J. Cancer* 145 (8), 2209–2224. <https://doi.org/10.1002/ijc.32338>.
- Dai, Y., Ai, B., Liu, Y., Pascal, L.E., Wang, Z., Dhir, R., Sun, X., Jiang, Y., 2021. Abnormal expression of rab27b in prostatic epithelial cells of benign prostatic hyperplasia alters intercellular communication. *Int. J. Biochem. Cell Biol.* 131, 105898. <https://doi.org/10.1016/j.biocel.2020.105898>.
- Eng, C., Jacome, A.A., Agarwal, R., Hayat, M.H., Byndloss, M.X., Holowatyj, A.N., Bailey, C., Lieu, C.H., 2022. A comprehensive framework for early-onset colorectal cancer research. *Lancet Oncol.* 23 (3), e116–e128. [https://doi.org/10.1016/S1470-2045\(21\)00588-X](https://doi.org/10.1016/S1470-2045(21)00588-X).
- Gou, H., Su, H., Liu, D., Wong, C.C., Shang, H., Fang, Y., Zeng, X., Chen, H., Li, Y., Huang, Z., Fan, M., Wei, C., Wang, X., Zhang, X., Li, X., Yu, J., 2023. Traditional medicine pientze huang suppresses colorectal tumorigenesis through restoring gut microbiota and metabolites. *Gastroenterology* 165 (6), 1404–1419. <https://doi.org/10.1053/j.gastro.2023.08.052>.
- Grothey, A., Fakih, M., Tabernero, J., 2021. Management of BRAF-Mutant metastatic colorectal cancer: a review of treatment options and evidence-based guidelines. *Ann. Oncol.* 32 (8), 959–967. <https://doi.org/10.1016/j.annonc.2021.03.206>.
- Guo, B., Zhuang, T., Li, C., Li, F., Shan, S., Zheng, M., Xu, Q., Wang, Y., Lei, L., Tang, K., Ouyang, W., Duan, J., Wu, Y., Cao, Y., Ullah, M.H.E., Zhou, Z., Lin, X., Wu, F., Xu, F., Liao, X., Yuan, L., 2024. MiRNA-132/212 encapsulated by adipose tissue-derived exosomes worsen atherosclerosis progression. *Cardiovasc. Diabetol.* 23 (1), 331. <https://doi.org/10.1186/s12933-024-02404-x>.
- Hendrix, A., Maynard, D., Pauwels, P., Braems, G., Denys, H., Van den Broecke, R., Lambert, J., Van Belle, S., Cocquyt, V., Gespach, C., Bracke, M., Seabra, M.C., Gahl, W.A., De Wever, O., Westbroek, W., 2010. Effect of the secretory small GTPase rab27b on breast cancer growth, invasion, and metastasis. *J. Natl. Cancer Inst.* 102 (12), 866–880. <https://doi.org/10.1093/jnci/djq153>.
- Jerchel, I.S., Hoogkamer, A.Q., Aries, I.M., Steeghs, E.M.P., Boer, J.M., Besselink, N.J.M., Boeree, A., van de Ven, C., de Groot-Kruseman, H.A., de Haas, V., Horstmann, M.A., Escherich, G., Zwaan, C.M., Cuppen, E., Koudijs, M.J., Pieters, R., den Boer, M.L., 2018. RAS pathway mutations as a predictive biomarker for treatment adaptation in pediatric b-cell precursor acute lymphoblastic leukemia. *Leukemia* 32 (4), 931–940. <https://doi.org/10.1038/leu.2017.303>.
- Jiang, Y., Guo, K., Wang, P., Zhu, Y., Huang, J., Ruan, S., 2022. The antitumor properties of atractylenolides: molecular mechanisms and signaling pathways. *Biomed. Pharmacother.* 155, 113699. <https://doi.org/10.1016/j.biopha.2022.113699>.
- Kang, Y., Chung, H., Nam, J., Park, H.J., Seo, E.K., Kim, Y.S., Lee, D., Lee, S.K., 2011. Cytotoxic and antineoplastic activity of timosaponin a-III for human colon cancer cells. *J. Nat. Prod.* 74 (4), 701–706. <https://doi.org/10.1021/np1007735>.
- Kanth, P., Inadomi, J.M., 2021. Screening and prevention of colorectal cancer. *Bmj* 374, n1855. <https://doi.org/10.1136/bmj.n1855>.
- Kitada, T., Asakawa, S., Hattori, N., Matsumine, H., Yamamura, Y., Minoshima, S., Yokochi, M., Mizuno, Y., Shimizu, N., 1998. Mutations in the parkin gene cause autosomal recessive juvenile parkinsonism. *Nature* 392 (6676), 605–608. <https://doi.org/10.1038/33416>.
- Koh, H.M., Jang, B.G., Kim, D.C., 2020. Prognostic significance of rab27 expression in solid cancer: a systematic review and meta-analysis. *Sci. Rep.* 10 (1), 14136. <https://doi.org/10.1038/s41598-020-71104-9>.
- Lemasters, J.J., 2014. Variants of mitochondrial autophagy: types 1 and 2 mitophagy and micromitophagy (type 3). *Redox Biol.* 2, 749–754. <https://doi.org/10.1016/j.redox.2014.06.004>.
- Li, A., Gao, M., Liu, B., Qin, Y., Chen, L., Liu, H., Wu, H., Gong, G., 2022. Mitochondrial autophagy: molecular mechanisms and implications for cardiovascular disease. *Cell Death Dis.* 13 (5), 444. <https://doi.org/10.1038/s41419-022-04906-6>.
- Li, M., Hou, X., Zhang, J., Wang, S., Fu, Q., He, L., 2011. Applications of HPLC/MS in the analysis of traditional chinese medicines. *J. Pharm. Anal.* 1 (2), 81–91. [https://doi.org/10.1016/S2095-1779\(11\)70015-6](https://doi.org/10.1016/S2095-1779(11)70015-6).
- Li, Y., Zheng, H., Luo, Y., Lin, Y., An, M., Kong, Y., Zhao, Y., Yin, Y., Ai, L., Huang, J., Chen, C., 2023. An HGF-dependent positive feedback loop between bladder cancer cells and fibroblasts mediates lymphangiogenesis and lymphatic metastasis. *Cancer Commun.* 43 (12), 1289–1311. <https://doi.org/10.1002/cac2.12470>.
- Lin, L., Zhou, X., Gao, T., Zhu, Z., Qing, Y., Liao, W., Lin, W., 2024. Herb pairs containing curcuma rhizoma (ezhu): a review of bio-active constituents, compatibility effects and t-copula function analysis. *J. Ethnopharmacol.* 319 (Pt 3), 117199. <https://doi.org/10.1016/j.jep.2023.117199>.
- Liu, L., Zhao, M., Lang, X., Jia, S., Kang, X., Liu, Y., Liu, J., 2024. Modified lichong decoction intervenes in colorectal cancer by modulating the intestinal flora and the wnt/beta-catenin signaling pathway. *J. Cancer Res. Clin. Oncol.* 150 (5), 234. <https://doi.org/10.1007/s00432-024-05763-w>.
- Lou, G., Palikaras, K., Lautrup, S., Scheibye-Knudsen, M., Tavernarakis, N., Fang, E.F., 2020. Mitophagy and neuroprotection. *Trends Mol. Méd. Sur* 26 (1), 8–20. <https://doi.org/10.1016/j.molmed.2019.07.002>.
- Lu, Y., Li, Z., Zhang, S., Zhang, T., Liu, Y., Zhang, L., 2023. Cellular mitophagy: mechanism, roles in diseases and small molecule pharmacological regulation. *Theranostics* 13 (2), 736–766. <https://doi.org/10.7150/thno.79876>.
- Luan, F., Ji, Y., Peng, L., Liu, Q., Cao, H., Yang, Y., He, X., Zeng, N., 2021. Extraction, purification, structural characteristics and biological properties of the polysaccharides from codonopsis pilosula: a review. *Carbohydr. Polym.* 261, 117863. <https://doi.org/10.1016/j.carbpol.2021.117863>.
- Luan, Y., Li, Y., Zhu, L., Zheng, S., Mao, D., Chen, Z., Cao, Y., 2018. Codonopsis bulleynana forest ex diels inhibits autophagy and induces apoptosis of colon cancer cells by activating the NF-kappaB signaling pathway. *Int. J. Mol. Med.* 41 (3), 1305–1314. <https://doi.org/10.3892/ijmm.2017.3337>.
- Moustapha, A., Peretout, P.A., Rainey, N.E., Sureau, F., Geze, M., Petit, J., Dewailly, E., Slomianny, C., Petit, P.X., 2015. Curcumin induces crosstalk between autophagy and apoptosis mediated by calcium release from the endoplasmic reticulum, lysosomal destabilization and mitochondrial events. *Cell Death Discov.* 1, 15017. <https://doi.org/10.1038/cddiscovery.2015.17>.
- Murphy, C.C., Zaki, T.A., 2024. Changing epidemiology of colorectal cancer - birth cohort effects and emerging risk factors. *Nat. Rev. Gastroenterol. Hepatol.* 21 (1), 25–34. <https://doi.org/10.1038/s41575-023-00841-9>.
- Narendra, D.P., Youle, R.J., 2024. The role of PINK1-parkin in mitochondrial quality control. *Nat. Cell Biol.* 26 (10), 1639–1651. <https://doi.org/10.1038/s41556-024-01513-9>.
- Nezich, C.L., Wang, C., Fogel, A.I., Youle, R.J., 2015. MIT/TFE transcription factors are activated during mitophagy downstream of parkin and atg5. *J. Cell Biol.* 210 (3), 435–450. <https://doi.org/10.1083/jcb.201501002>.
- Ni, Q., Zhong, R., Yang, Y., Li, G., Yang, W., Zhou, J., Hu, Y., Wu, J., Zhu, Y., 2025. Dan'e fukang decoction reduces hemorrhage in a rat model of mifepristone induced incomplete abortion and may correlate with cell adhesion molecule signaling interference. *J. Ethnopharmacol.* 337 (Pt 3), 118984. <https://doi.org/10.1016/j.jep.2024.118984>.
- Okamoto, K., 2014. Organellaphagy: eliminating cellular building blocks via selective autophagy. *J. Cell Biol.* 205 (4), 435–445. <https://doi.org/10.1083/jcb.201402054>.
- Ostrowski, M., Carmo, N.B., Krumeich, S., Fanget, I., Raposo, G., Savina, A., Moita, C.F., Schauer, K., Hume, A.N., Freitas, R.P., Goud, B., Benaroch, P., Hacohen, N., Fukuda, M., Desnos, C., Seabra, M.C., Darchen, F., Amigorena, S., Moita, L.F., Thery, C., 2010. Rab27a and rab27b control different steps of the exosome secretion pathway. *Nat. Cell Biol.* 12 (1), 19–30. <https://doi.org/10.1038/ncb2000>.
- Picca, A., Faltz, J., Auwerx, J., Ferrucci, L., D'Amico, D., 2023. Mitophagy in human health, ageing and disease. *Nat. Metab.* 5 (12), 2047–2061. <https://doi.org/10.1038/s42255-023-00930-8>.
- Ren, J., Xing, B., Lv, K., O'Keefe, R.A., Wu, M., Wang, R., Bauer, K.M., Ghazaryan, A., Burslem, G.M., Zhang, J., O'Connell, R.M., Pillai, V., Hexner, E.O., Phillips, M.R., Tong, W., 2023. RAB27b controls palmitoylation-dependent NRAS trafficking and signaling in myeloid leukemia. *J. Clin. Investig.* 133 (12). <https://doi.org/10.1172/JCI165510>.

- Shi, J., Fung, G., Piesik, P., Zhang, J., Luo, H., 2014. Dominant-negative function of the c-terminal fragments of NBR1 and SQSTM1 generated during enteroviral infection. *Cell Death Differ.* 21 (9), 1432–1441. <https://doi.org/10.1038/cdd.2014.58>.
- Sun, R., Sun, C., Yue, Z., Yin, G., Zhou, L., Zhang, S., Zhang, Y., Tang, D., Tan, X., 2024. Astragali radix-curcumae rhizoma herb pair reduces the stemness of colorectal cancer cells through HIF-2alpha/beta-catenin pathway. *Phytomedicine* 132, 155824. <https://doi.org/10.1016/j.phymed.2024.155824>.
- Tang, D., Xu, X., Ying, J., Xie, T., Cao, G., 2020. Transfer of metastatic traits via mir-200c in extracellular vesicles derived from colorectal cancer stem cells is inhibited by atractylenolide i. *Clin. Transl. Med.* 10 (4), e139. <https://doi.org/10.1002/ctm2.139>.
- Underwood, R., Wang, B., Carico, C., Whitaker, R.H., Placzek, W.J., Yacoubian, T.A., 2020. The GTPase rab27b regulates the release, autophagic clearance, and toxicity of alpha-synuclein. *J. Biol. Chem.* 295 (23), 8005–8016. <https://doi.org/10.1074/jbc.RA120.013337>.
- Van Humbeeck, C., Cornelissen, T., Hofkens, H., Mandemakers, W., Gevaert, K., De Strooper, B., Vandenbergh, W., 2011. Parkin interacts with ambra1 to induce mitophagy. *J. Neurosci.* 31 (28), 10249–10261. <https://doi.org/10.1523/JNEUROSCI.1917-11.2011>.
- Wang, J., Pu, X., Zhuang, H., Guo, Z., Wang, M., Yang, H., Li, C., Chang, X., 2024. Astragaloside IV alleviates septic myocardial injury through DUSP1-prohibitin 2 mediated mitochondrial quality control and ER-autophagy. *J. Adv. Res.* <https://doi.org/10.1016/j.jare.2024.10.030>.
- Wang, Q., Lu, W., Yin, T., Lu, L., 2019. Calycosin suppresses TGF-beta-induced epithelial-to-mesenchymal transition and migration by upregulating BATF2 to target PAI-1 via the wnt and PI3k/akt signaling pathways in colorectal cancer cells. *J. Exp. Clin. Cancer Res.* 38 (1), 240. <https://doi.org/10.1186/s13046-019-1243-7>.
- Wei, X., Leng, X., Liang, J., Liu, J., Chi, L., Deng, H., Sun, D., 2024. Pharmacological potential of natural medicine astragali radix in treating intestinal diseases. *Biomed. Pharmacother.* 180, 117580. <https://doi.org/10.1016/j.biopha.2024.117580>.
- Wei, Y., Fan, Y., Huang, S., Lv, J., Zhang, Y., Hao, Z., 2024. Baizhu shaoyao decoction restores the intestinal barrier and brain-gut axis balance to alleviate diarrhea-predominant irritable bowel syndrome via FoxO1/FoxO3a. *Phytomedicine* 122, 155163. <https://doi.org/10.1016/j.phymed.2023.155163>.
- Xiao, X., Wang, W., Li, Y., Yang, D., Li, X., Shen, C., Liu, Y., Ke, X., Guo, S., Guo, Z., 2018. HSP90AA1-mediated autophagy promotes drug resistance in osteosarcoma. *J. Exp. Clin. Cancer Res.* 37 (1), 201. <https://doi.org/10.1186/s13046-018-0880-6>.
- Xu, N., Han, X., Zhang, X., Wang, J., Yuan, J., Wang, M., Wu, H., Huang, F., Shi, H., Yang, L., Wu, X., 2024. Huangqi-guizhi-wuwu decoction regulates differentiation of CD4(+) t cell and prevents against experimental autoimmune encephalomyelitis progression in mice. *Phytomedicine* 125, 155239. <https://doi.org/10.1016/j.phymed.2023.155239>.
- Young, E., Rajagopalan, A., Tee, D., Sathananthan, D., Hoile, S., Singh, R., 2024. Texture and color enhancement imaging improves colonic adenoma detection: a multicenter randomized controlled trial. *Gastroenterology* 166 (2), 338–340. <https://doi.org/10.1053/j.gastro.2023.10.008>.
- Zhang, Y., Ding, X., Zhang, Q., Zeng, C., Chen, H., Lu, L., 2024. Trichosanthin elicits antitumor activity via MICU3 mediated mitochondria calcium influx. *J. Adv. Res.* <https://doi.org/10.1016/j.jare.2024.11.001>.
- Zou, Y., Wang, S., Zhang, H., Gu, Y., Chen, H., Huang, Z., Yang, F., Li, W., Chen, C., Men, L., Tian, Q., Xie, T., 2024. The triangular relationship between traditional chinese medicines, intestinal flora, and colorectal cancer. *Med. Res. Rev.* 44 (2), 539–567. <https://doi.org/10.1002/med.21989>.